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Department of Computer Science and Engineering
DATABASE MANAGEMENT SYSTEMS (CD252IA)

Synopsis

TITLE: ConnectomeChronicle - A Temporal, Federated and Distributed Database for the Evolution of Neural Connectomes		
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1. Introduction

ConnectomeChronicle is a database-first framework for integrating and managing heterogeneous, evolving neural connectome datasets. It integrates datasets such as MaleCNS, FlyWire, hemibrain, and MANC through source-specific pipelines, harmonizes their schemas into a canonical representation while preserving source identifiers and provenance, and supports temporal storage, entity resolution, graph comparison, and distributed querying, with AI/ML assisting entity matching and anomaly detection. Modern connectomes are large-scale graphs; for example, the whole-brain female *Drosophila* connectome contains approximately 139,255 neurons and 54.5 million synapses.

2. Existing System

Existing connectome platforms address distinct aspects of data management: CAVE supports version-aware annotation and point-in-time querying, neuPrint provides graph-based storage and querying, and FlyBrainLab/NeuroArch focuses on heterogeneous data integration. ConnectomeChronicle builds on these capabilities by investigating their integration within a unified DBMS architecture supporting schema harmonization, source-preserving entity resolution, temporal and provenance-aware storage, graph differencing, distributed partitioning, and query processing.

3. Proposed System

ConnectomeChronicle provides a unified DBMS-oriented architecture in which connectome datasets are ingested through source-specific ETL pipelines while preserving original identifiers and metadata, then transformed into a canonical schema comprising entities such as CanonicalNeuron, NeuronType, BrainRegion, Connection, and DatasetRelease. An explicit entity-resolution layer records source-to-canonical correspondences with mapping methods, confidence, and evidence, while uncertain or unresolved mappings are retained. Temporal or, where feasible, bitemporal storage preserves historical states of neurons, connections, and annotations, with provenance tracking their source datasets, releases, and transformations.

ConnectomeDiff identifies additions, deletions, and modifications across releases, with Isolation Forest used to flag statistically unusual changes. For scalability, the system uses PostgreSQL/Docker-based shards and evaluates identifier-, anatomical-region-, and connectivity-community-based partitioning using query latency, cross-shard communication, load balance, and distributed join overhead, making integration, temporal management, provenance, graph comparison, and distributed querying core database operations.

4. Relational Database Structure

The relational schema consists of DATASET, RELEASE, DATASET_NEURON, CANONICAL_NEURON, NEURON_MAPPING, CONNECTION, ANNOTATION, and PROVENANCE, with relationships defining how these entities interact. A DATASET has multiple RELEASEs and PROVENANCE records, while each RELEASE defines multiple DATASET_NEURON records, contains CONNECTIONs, provides ANNOTATIONs, and is associated with PROVENANCE. DATASET_NEURON records are mapped by NEURON_MAPPING to CANONICAL_NEURON entities, with mapping confidence and evidence retained. CANONICAL_NEURON entities send and receive CONNECTIONs, have ANNOTATIONs, and are referenced by mappings, while PROVENANCE records trace source files, records, transformations, and pipeline versions. These relationships preserve dataset versioning, source-to-canonical mapping, connectivity, annotations, temporal information, and end-to-end data provenance.

5. RDBMS and NoSQL Integration

The project adopts a hybrid RDBMS–NoSQL architecture where PostgreSQL maintains the authoritative structured state, including canonical entities, releases, mappings, temporal records, connections, provenance, and partition metadata, with relational constraints and transactions ensuring consistency. MongoDB stores heterogeneous, source-specific information such as annotations, morphology, experimental metadata, and semi-structured evidence. The two stores are linked through common source and canonical identifiers, combining relational consistency with NoSQL flexibility.

6. Societal Concern

The project supports reproducible and responsible neuroscience research by preserving historical states, provenance, and uncertain entity mappings rather than treating uncertain correspondences as definitive. Graph embeddings and GraphSAGE assist in generating cross-dataset neuron correspondences, while Isolation Forest identifies unusual connectivity or annotation changes; the database remains responsible for authoritative storage, validation, provenance, and temporal management.